

Article

A Novel Nonsingular Fast Terminal Sliding Mode Control with Sliding Mode Disturbance Observer for Permanent Magnet Synchronous Motor Servo Control

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Abstract: This article proposes a novel nonsingular fast terminal sliding mode control (N-NFTSMC) with a sliding mode disturbance observer (SDOB) for permanent magnet synchronous motor (PMSM) servo control. Firstly, to reduce the chattering issue, a new sliding mode reaching law (NSRL) is proposed for the N-NFTSMC. Secondly, to further improve the dynamic tracking accuracy, we introduce a sliding disturbance observer to estimate unknown disturbances for feedforward compensation. Comparative simulations via Matlab/Simulink 2018 are conducted using the traditional NFTSMC and N-NFTSMC; the step simulation results show that the chattering phenomenon was suppressed well via the N-NFTSMC scheme. The sine wave tracking simulation proves that the N-NFTSMC has better dynamic tracking performance when compared with traditional NFTSMC. Finally, we carry out experiments to validate that the N-NFTSMC adequately suppresses the chattering issue and possesses better anti-disturbance performance.

Keywords: PMSM; nonsingular terminal sliding mode control; sliding mode reaching law; sliding mode disturbance observer



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1. Introduction

Permanent magnet synchronous motors (PMSMs) have been applied in many AC servo systems due to their simple structure and fast response time [1–3]. Therefore, PMSMs are widely used in modern industrial fields like robot manipulators, pneumatic servo systems [4], electric vehicles (EVs) and spacecraft. The traditional proportional–integral–derivative (PID) controller is usually used in PMSMs because its parameters are easily adjustable [5]. However, the PMSM system contains many unknown dynamics that will worsen the control performance of the PMSM. The PID controller has difficulties meeting the requirements needed for the high-precision control performance of PMSMs. Therefore, it is important to develop advanced control methods to deal with these problems. To tackle these issues, plenty of novel control methods have been proposed, for example, active disturbance control (ADRC) [6], sliding mode control (SMC) [7] and model predictive control [8].

Among these approaches, SMC has a performance advantage due to its robustness [9–13]. However, there are still some problems that need to be solved. The system error for the typical linear SMC is not capable of finite-time convergence, and the chattering phenomenon must be suppressed. Y. Sun's novel nonsingular terminal sliding mode control (NTSMC) is an approach that guarantees system stabilization within a bounded time interval [14]. A full-order nonsingular terminal sliding mode control method is proposed in [15] for dynamic

systems subject to both matched and mismatched disturbances. The chattering problem may cause negative effects and destabilizes the system in real applications [16]. Different strategies have been developed to reduce the chattering phenomenon, such as high-order SMC [17,18] and advanced sliding mode reaching law in [19–23]. Among these methods, the advanced sliding mode reaching law can decrease the chattering issue effectively. It is contradictory to overcome heavy chattering and obtain a fast response at same time. Therefore, an adaptive neural network nonsingular fast terminal sliding mode control (ANNNFTSMC), which has advantages that include a fast response speed and a small static error, is investigated in [24].

To enhance the anti-disturbance performance, the disturbance observer has been widely studied [25–27]. A time-varying nonlinear disturbance observer (TVNDO), as specified in [28], was designed to evaluate unknown disturbances. There are many nonlinear disturbances in practical engineering applications; however, PI control cannot meet the demands of dynamic performance. Based on this problem, an improved model-free continuous super-twisting NFTSM is applied for IPMSM in [29]. Considering that disturbances are usually changing in engineering environments, a DOB, which deals with time-varying disturbances effectively, is designed and analyzed in [30,31].

Although NFTSMC has a high convergent speed, it still has chattering issues. Based on the analysis above, an effective way to decrease the chattering issue is to modify the sliding mode reaching law. We have devised an NSMRL to reduce chattering. When applied to the motion control of PMSMs, the tracking accuracy under complicated working conditions can be improved. In this paper, we propose a novel NFTSMC with a sliding mode observer (SDOB); the control structure can be seen in Figure 1. The contribution of this study can be summarized as follows:

- (1) To reduce the chattering problem in the traditional NFTSMC, an NSMRL is designed for the N-NFTSMC. The NSMRL with a smooth switch function eliminates chattering behavior effectively.
- (2) An SDOB is proposed to estimate the unknown disturbances of the PMSM system. The N-NFTSMC with the SDOB is designed to improve the anti-disturbance performance of the system.
- (3) Simulation and experimental verification are conducted to prove the effectiveness of the N-NFTSMC and to compare it with the classic NFTSMC.

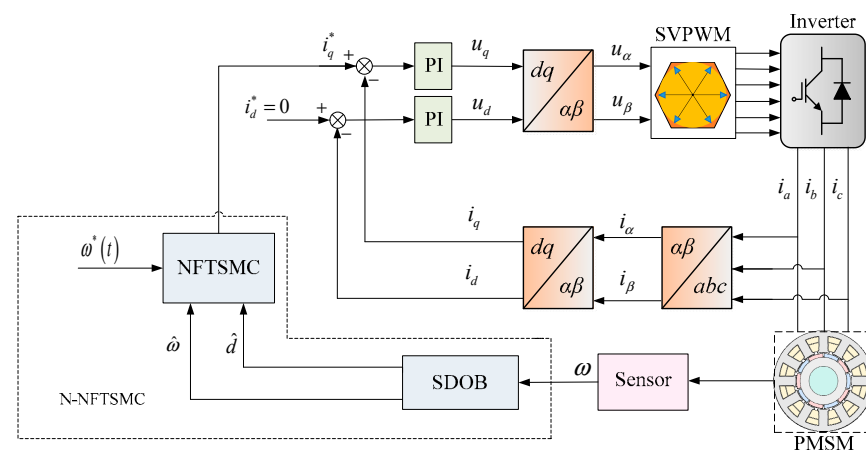


Figure 1. Control structure of the PMSM.

The remainder of this paper is organized as follows: Section 2 states the traditional NFTSMC design and problem formulation. Section 3 states the proposed N-NFTSMC. Section 4 details the design of the SDOB and its convergence analysis. Section 5 verifies the effectiveness of the proposed control method using a Simulink simulation. Section 6 describes experiments to validate the proposed N-NFTSMC.

2. NFTSMC Design

2.1. Modelling of PMSM

Based on the theoretical speculations regarding the control of permanent magnet synchronous motors (PMSMs), with the d-axis current set at 0, the d–q axis mathematical model is presented as follows:

$$\begin{cases} \dot{i}_q = -\frac{R}{L}i_q - n\omega i_d - \frac{n\psi_f}{L} + \frac{u_q}{L} \\ \dot{\omega} = \frac{1}{J}(K_e i_q - T_L - B\omega) \end{cases} \quad (1)$$

where u_q is the stator voltage; i_d and i_q denote the currents within the stator; L is the stator inductance; ψ_f stands for the magnetic flux; R signifies the stator resistance; n denotes the pole-pair counts of the PMSM; ω is rotational speed; J indicates the rotor's inertial moment; K_e represents the motor's torque coefficient ($T_e = \frac{3}{2}n\psi_f i_q = K_e i_q$); T_L represents the torque applied by the load; and B denotes the friction coefficient.

2.2. Definition of State Variables

If the reference speed is defined as ω^* and the real speed is defined as ω , then the error between ω^* and ω can be obtained by

$$\begin{cases} e = \omega^* - \omega \\ \dot{e} = -\dot{\omega} \end{cases} \quad (2)$$

In the experimental verification, T_L is also set at a constant value, which can be obtained by $\dot{T}_L = 0$. Then, the derivative of Equation (2) is given as

$$\begin{bmatrix} \dot{e} \\ \ddot{e} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{b_m}{J} \end{bmatrix} \begin{bmatrix} \dot{e} \\ \ddot{e} \end{bmatrix} + \begin{bmatrix} 0 \\ -\frac{1.5p_n\psi}{J} \end{bmatrix} \dot{i}_q \quad (3)$$

The state variables can be summarized as

$$\begin{bmatrix} e \\ \dot{e} \\ \ddot{e} \end{bmatrix} = \begin{bmatrix} \omega^* - \omega \\ -\frac{1.5p_n\psi}{J}i_q + \frac{b_m}{J}\omega + \frac{T_L}{J} \\ -\frac{1.5p_n\psi}{J}\dot{i}_q + \frac{b_m}{J}\dot{\omega} \end{bmatrix} \quad (4)$$

2.3. Classic Nonsingular Fast Terminal Sliding Mode Control

The classic nonsingular fast terminal sliding mode surface is set forth as

$$s = e + k_1|e|^{\sigma_1}\text{sign}(e) + k_2|\dot{e}|^{\sigma_2}\text{sign}(\dot{e}) \quad (5)$$

The appropriate sliding mode reaching law ensures that the system variable states convergence to the designed sliding surface in a finite time. The exponential sliding mode control law is similarly formulated as follows:

$$\dot{s} = -\varepsilon\text{sign}(s) - ks, \quad \varepsilon > 0, k > 0. \quad (6)$$

The derivation of the sliding mode surface of (5) can be written as:

$$\begin{aligned} \dot{s} &= \dot{e} + \sigma_1 k_1 |e|^{\sigma_1-1} \text{sign}(e) \dot{e} + k_2 \sigma_2 |\dot{e}|^{\sigma_2-1} \text{sign}(\dot{e}) \\ &= \left(1 + \sigma_1 k_1 |e|^{\sigma_1-1} \text{sign}(e)\right) \dot{e} + k_2 \sigma_2 |\dot{e}|^{\sigma_2-1} \text{sign}(\dot{e}) \end{aligned} \quad (7)$$

Combined with (4), (5) and (7), the output of the classic NFTSMC is generated according to the subsequent formula:

$$\dot{i}_q = \frac{J}{1.5p_n\psi} \cdot \frac{1}{k_2\sigma_2|\dot{e}|^{\sigma_2-1}\text{sign}(\dot{e})} \int_0^t \left[\left(1 + \sigma_1 k_1 |e|^{\sigma_1-1} \text{sign}(e) - \frac{b_m}{J} \sigma_2 k_2 |\dot{e}|^{\sigma_2-1} \text{sign}(\dot{e})\right) \dot{e} + \varepsilon \text{sign}(s) + ks \right] dt \quad (8)$$

3. N-NFTSMC Design

The overall block diagram of the N-NFTSMC controller is shown in Figure 2. It depicts the connections between the mathematical models.

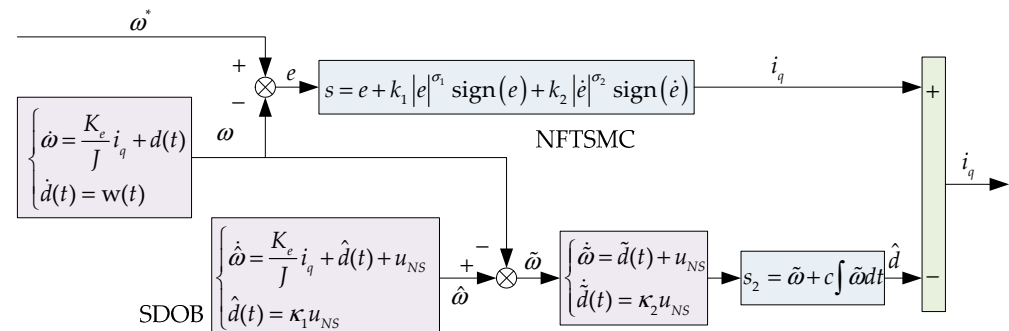


Figure 2. The block diagram of N-NFTSMC.

3.1. Controller Design

Equation (6) indicates that an increase in the coefficients leads to an improved system convergence speed. On the other hand, the process of convergence will also cause chattering. Therefore, it is difficult to maintain the balance between the convergence speed and the amount of chattering using the traditional NFTSMC. The main reason is that the reaching law includes a switching function $\text{sign}(\cdot)$. To cope with this issue, we replace $\text{sign}(\cdot)$ with $\tanh(\cdot)$, which can provide a smooth transition. A comparison of the two functions is shown in Figure 3.

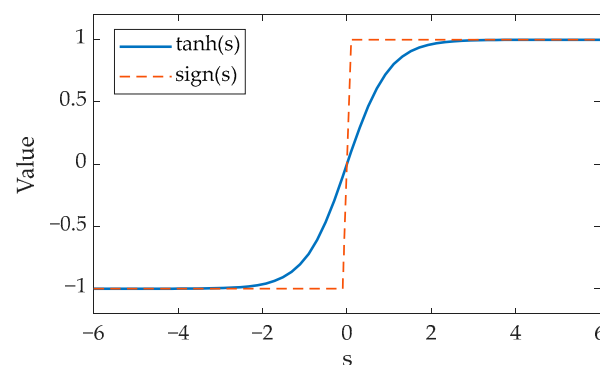


Figure 3. Function $\tanh(s)$ and $\text{sign}(s)$.

To reduce the chattering, a new sliding mode reaching law (NSMRL) based on the previously mentioned sliding mode reaching law in Equation (6) is designed as follows:

$$\begin{cases} \dot{s} = -\varepsilon_1 \frac{s}{1+|s|} - \varepsilon_2 |s|^{\alpha_1} \tanh(s) - \varepsilon_3 |s|^{\alpha_2} \tanh(s) \\ \varepsilon_1 > 0, \varepsilon_2 > 0, \varepsilon_3 > 0, 0 < \alpha_1 < 1, \alpha_2 > 1. \end{cases} \quad (9)$$

It can be learned from Equation (9) that when the system state is far away from the sliding mode surface s , the terms $\varepsilon_1 \frac{s}{1+|s|}$ and $\varepsilon_3 |s|^{\alpha_2} \tanh(s)$ will accelerate the state convergence to the sliding mode. When the state is close to the sliding mode surface, the term $\varepsilon_2 |s|^{\alpha_1} \tanh(s)$ works and makes the states reach the sliding mode surface. In Figure 4, there are three sets of reaching laws. As depicted in Figure 4, the red line, which denotes the NSMRL, achieves convergence with the greatest speed despite being distant from equilibrium.

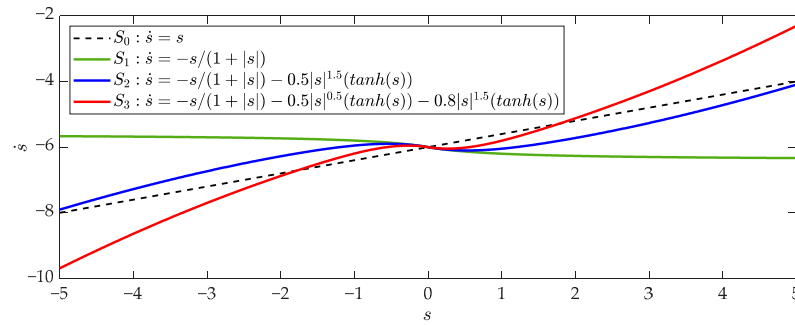


Figure 4. Comparison of the variants of the different sliding mode reaching laws.

The NSMRL in (9) is equal to the derivation of (5), combined with (7). The control law of the N-NFTSMC is presented below:

$$i_q = \frac{J}{1.5p_n\psi} \cdot \frac{1}{k_2\sigma_2|\dot{e}|^{\sigma_2-1}\text{sign}(\dot{e})} \int_0^t \left[\left(1 + \sigma_1 k_1 |e|^{\sigma_1-1} \text{sign}(e) - \frac{b_m}{J} \sigma_2 k_2 |\dot{e}|^{\sigma_2-1} \text{sign}(\dot{e}) \right) \dot{e} + \varepsilon_1 \frac{s}{1+|s|} + \varepsilon_2 |s|^{\alpha_1} \tanh(s) + \varepsilon_3 |s|^{\alpha_2} \tanh(s) \right] dt \quad (10)$$

3.2. Stability Analysis for Novel NFTSMC

The Lyapunov function Equation (11) is used to analyze the stability of the N-NFTSMC:

$$V_1 = \frac{1}{2} s^2 \quad (11)$$

Combining (5), (9) and (10), the derivation of (11) can be obtained as

$$\begin{aligned} \dot{V}_1 &= s \left(\dot{e} + \sigma_1 k_1 |e|^{\sigma_1-1} \text{sign}(e) \dot{e} + k_2 \sigma_2 |\dot{e}|^{\sigma_2-1} \text{sign}(\dot{e}) \ddot{e} \right) \\ &= s \left\{ \dot{e} + \sigma_1 k_1 |e|^{\sigma_1-1} \text{sign}(e) \dot{e} + k_2 \sigma_2 |\dot{e}|^{\sigma_2-1} \text{sign}(\dot{e}) \left[-\frac{1}{k_2 \sigma_2 |\dot{e}|^{\sigma_2-1} \text{sign}(\dot{e})} \left(1 + \sigma_1 k_1 |e|^{\sigma_1-1} \text{sign}(e) \dot{e} - \frac{b_m}{J} k_2 \sigma_2 |\dot{e}|^{\sigma_2-1} \text{sign}(\dot{e}) - \dot{s} \right) + \frac{b_m}{J} \dot{\omega} \right] \right\} \\ &= s \left(-\varepsilon_1 \frac{s}{1+|s|} - \varepsilon_2 |s|^{\alpha_1} \tanh(s) - \varepsilon_3 |s|^{\alpha_2} \tanh(s) \right) \\ &= -\varepsilon_1 \frac{s^2}{1+|s|} - \varepsilon_2 |s|^{\alpha_1} s \cdot \tanh(s) - \varepsilon_3 |s|^{\alpha_2} s \cdot \tanh(s) < 0 \end{aligned} \quad (12)$$

4. Sliding Mode Disturbance Observer

To overcome the chattering problem with sliding mode control, this section proposes a new sliding mode disturbance observer. The convergence of the SDOB is proved in Section 4.2.

4.1. SDOB Design

By defining $d(t) = \frac{1}{J}(-T_L - B\omega)$, the mathematical model of PMSM (Equation (1)) can be simplified as

$$\begin{cases} \dot{\omega} = \frac{K_e}{J} i_q + d(t) \\ \dot{d}(t) = w(t) \end{cases} \quad (13)$$

According to the motion equation of the PMSM, we can model the sliding mode observer (SDOB) as follows:

$$\begin{cases} \dot{\hat{\omega}} = \frac{K_e}{J} i_q + \hat{d}(t) + u_{NS} \\ \dot{\hat{d}}(t) = \kappa_1 u_{NS} \end{cases} \quad (14)$$

Equation (14) minus (13) results in the error equation of the SDOB:

$$\begin{cases} \dot{\tilde{\omega}} = \tilde{d}(t) + u_{NS} \\ \dot{\tilde{d}}(t) = \kappa_2 u_{NS} \end{cases} \quad (15)$$

where $\kappa_2 = \kappa_1 - w(t)$. The sliding mode of the SDOB is designed as follows:

$$s_2 = \tilde{\omega} + c \int \tilde{\omega} dt \quad (16)$$

where $\tilde{\omega} = \hat{\omega} - \omega$ and $\tilde{d}(t) = \hat{d}(t) - d(t)$ represent the errors in the estimation of speed and disturbance, respectively. The course of the hyperbolic tangent over time is shown in Figure 2. The reaching law is thus designed as

$$\dot{s}_2 = -\lambda \tanh(s_2) \quad (17)$$

Combined with Equation (15), (16) and (17), u_{NS} can be obtained:

$$u_{NS} = -\lambda \tanh(s_2) - \tilde{d}(t) \quad (18)$$

4.2. Stability Analysis for SDOB

The Lyapunov function is used to prove the stability of the designed SDOB:

$$V_2 = \frac{1}{2} s_2^2 \quad (19)$$

After differentiating V_2 , we can obtain

$$\dot{V}_2 = s_2 \dot{s}_2 = s_2 (-\lambda \tanh(s_2)) \leq 0 \quad (20)$$

where $\lambda > 0$.

5. Simulation Verification

To verify the control performance of the proposed N-NFTSMC, simulation experiments are conducted in this section. Firstly, step response simulation experiments are conducted to verify whether the response time of the system is improved using the novel method. Secondly, sine wave tracking simulation is used to verify the proposed method.

5.1. Step Response Experiment

To verify the improved response time of the N-NFTSMC method, it is compared with the SMC method through a step response simulation. The simulation results can be seen in Figure 5. The SMC method converges to the reference signal at 0.016 s, while the novel NFTSMC method converges to the reference signal at 0.013 s. These results show that the N-NFTSMC method converges faster to the equilibrium state.

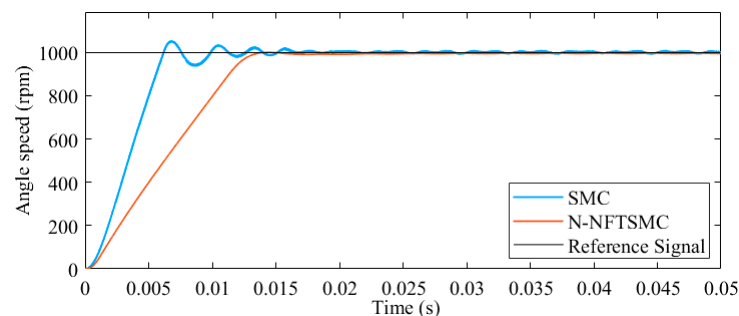
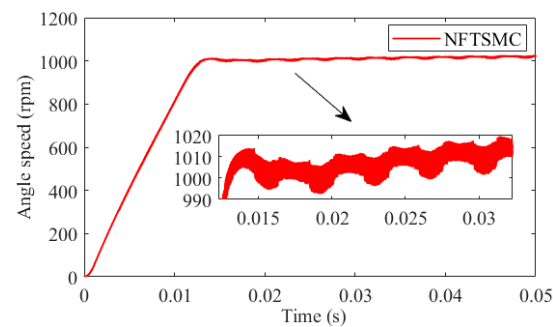
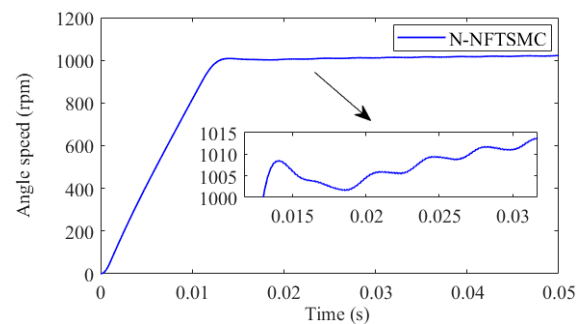


Figure 5. Comparison of the step responses for SMC and N-NFTSMC.

A step response simulation is conducted to verify the suppression of chattering using the N-NFTSMC proposed in this paper. The sliding mode surface parameters are set as $k_1 = 2$ and $k_2 = 0.8$ for the NFTSMC and N-NFTSMC. For the NFTSMC, the reaching law parameters are listed in Table 1. The NFTSMC uses the traditional reaching law (6), and the N-NFTSMC uses the NSMRL (9). The simulation results are shown in Figures 6 and 7.

Table 1. The sliding mode reaching law parameters.

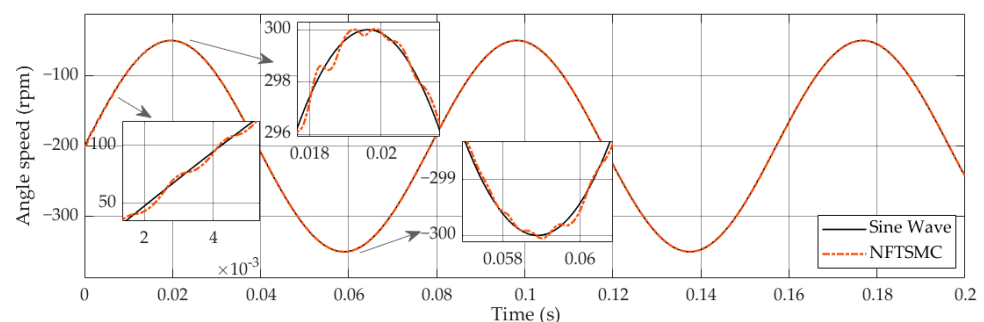
Parameter	Value
ε_1	400
ε_2	0
α_1	1/2
α_2	3/2

**Figure 6.** Step response by traditional NFTSMC.**Figure 7.** Step response by N-NFTSMC.

Chattering can be seen in Figure 6. Compared with the traditional NFTSMC, the chattering problem is solved by the N-NFTSMC, as shown in Figure 7. This proves the effectiveness of the NSMRL to suppress chattering.

5.2. Sinusoidal Tracking Simulation

To further verify the effectiveness of the N-NFTSMC approach, firstly, a sine wave tracking simulation is conducted in this section using the NFTSMC. A sine wave with an amplitude of 300 is set as the reference signal. The simulation results can be seen in Figure 8. The chattering phenomenon is more significant near the two poles of the sine curve.

**Figure 8.** Sine wave tracking by traditional NFTSMC.

Secondly, the same simulation experiment is repeated using the N-NFTSMC approach. The tracking curve using the N-NFTSMC is shown in Figure 9. The tracking curve of N-NFTSMC is smoother and closer to the sine wave compared with the NFTSMC curve.

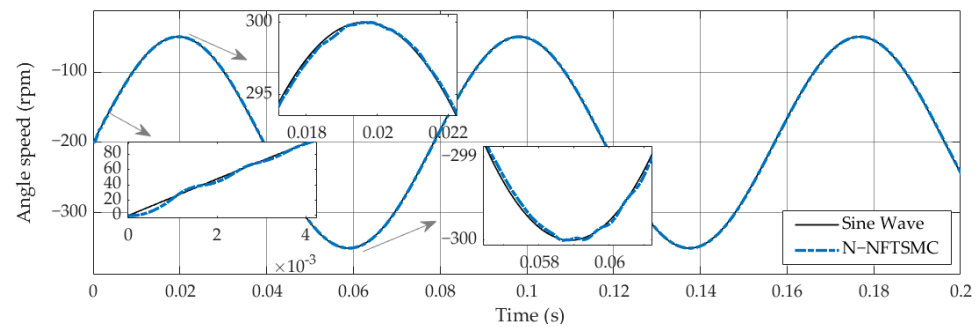


Figure 9. Sine wave tracking by N–NFTSMC.

To compare the simulation results of the two sets of experiments more intuitively, the tracking error curves are shown in Figure 10. The maximum error using the N-NFTSMC method is about 0.61 rpm, while the maximum error using the traditional NFTSMC method is about 0.98 rpm. The sine wave tracking simulation results show that the N-NFTSMC method can track the reference signal more accurately than the NFTSMC method.

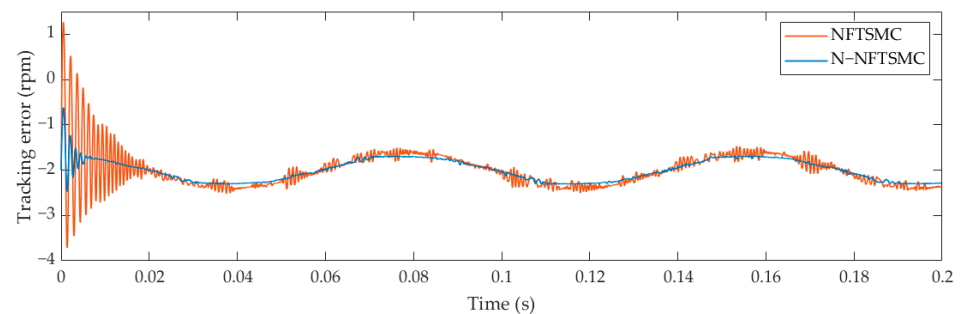


Figure 10. Sine wave tracking error.

The electromagnetic torque curve is also given in Figure 11. It is obvious that the blue line representing the N-NFTSMC is smoother. The sinusoidal tracking simulation result shows that the N-NFTSMC proposed in this paper has a better dynamic tracking performance than the traditional NFTSMC method.

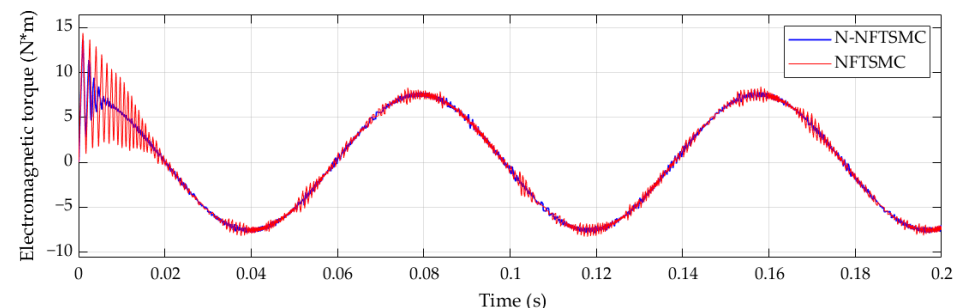


Figure 11. Electromagnetic torque curve.

6. Experimental Verification

To prove the competence of the N-NFTSMC method, semi-physical experiments are implemented on a PMSM servo platform, as portrayed in Figure 12.



1-Computer 2-Controller 3-Tension controller
 4-Driving circuit board 5-Current measurement circuits
 6-Inverter 7-PMSM 8-Position sensor 9-Magnetic particle brake

Figure 12. Experimental setup of PMSM servo system.

The PMSM in this servo system is a four-phase surface-mounted PMSM. The specialized parameters of the PMSM are detailed in Table 2.

Table 2. Parameters of the PMSM.

Parameter	Value
Rated power	200 W
Line resistance	0.33 Ω
Line inductance	9×10^{-4} H
Number of pole pairs	4
Torque coefficient	0.087 Nm/A
Rated voltage	36 V
Rated current	7.5 A

For experimental comparisons, the comparative transient response tests for the NFTSMC and N-NFTSMC control strategies are carried out at 500 rpm and 1000 rpm, respectively. The experimental results of the reference speed, 500 rpm, are shown in Figure 13a. It can be seen that when the motor runs at 500 rpm, after reaching the steady state, the maximum speed fluctuations based on the NFTSMC and N-NFTSMC are about 10 rpm and 6 rpm, respectively. Changing the reference speed to 1000 rpm increases the maximum speed fluctuations based on the NFTSMC and N-NFTSMC to 20 rpm and 12 rpm, as shown in Figure 13b. These results demonstrate that the N-NFTSMC reduces the chattering problem.

In the PMSM servo control, the d-axis is set at 0, and the q-axis is used for output torque control. The chattering of the q-axis voltage directly influences the velocity chattering. Figure 14 shows a comparison of the q-axis voltage using the NFTSMC and N-NFTSMC. With no load torque, the PMSM starts to speed up from 0 to 500 rpm and from 0 to 1000 rpm. It can be learned from the two sets of experiments that the q-axis voltage chattering range is smaller using the N-NFTSMC.

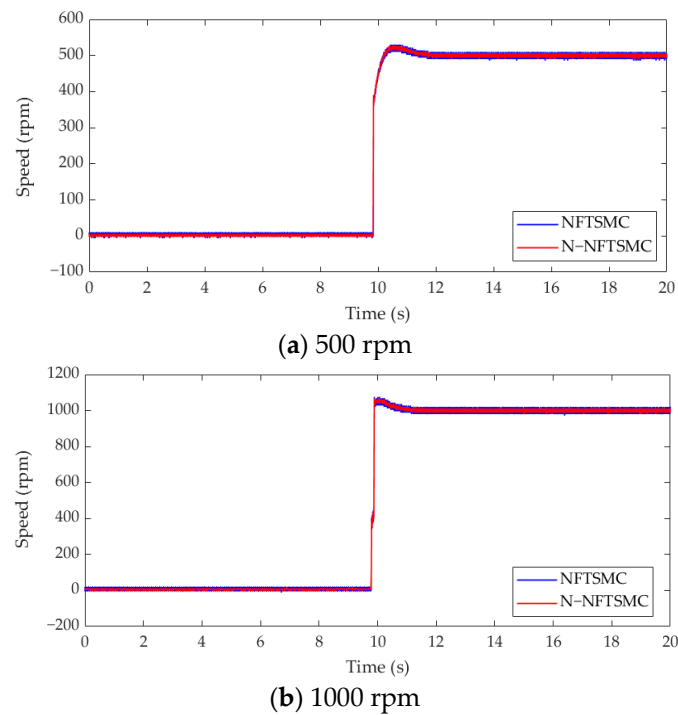


Figure 13. Startup responses of PMSM with NFTSMC and N-NFTSMC: (a) 500 rpm and (b) 1000 rpm.

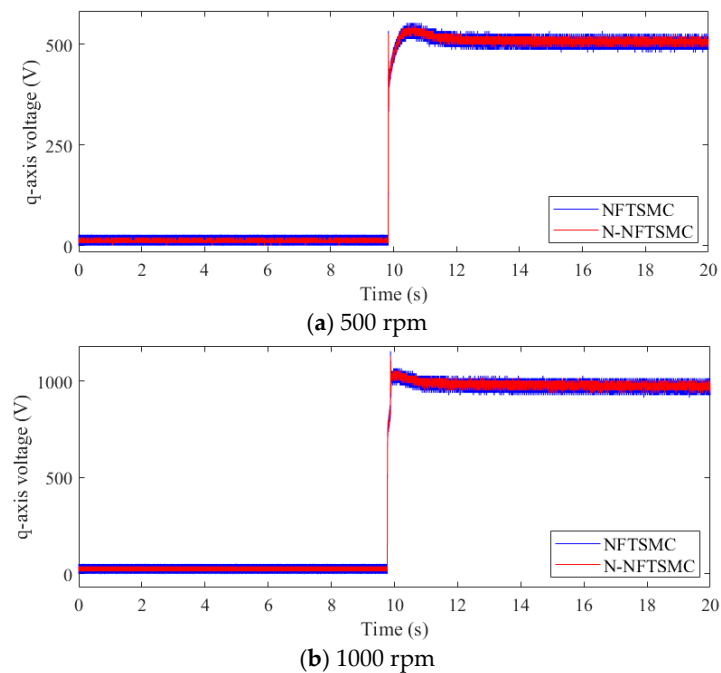


Figure 14. Q-axis voltage for step response: (a) 500 rpm and (b) 1000 rpm.

To demonstrate that the N-NFTSMC possesses better robustness than the NFTSMC, loading experiments are conducted under a speed command of 500 rpm. When the PMSM stably rotates at 500 rpm, we add a sudden load at 10 s using the magnetic powder brake and observe its influence on the PMSM velocities. As can be seen in Figure 15, the N-NFTSMC showed a better anti-disturbance performance than the NFTSMC.

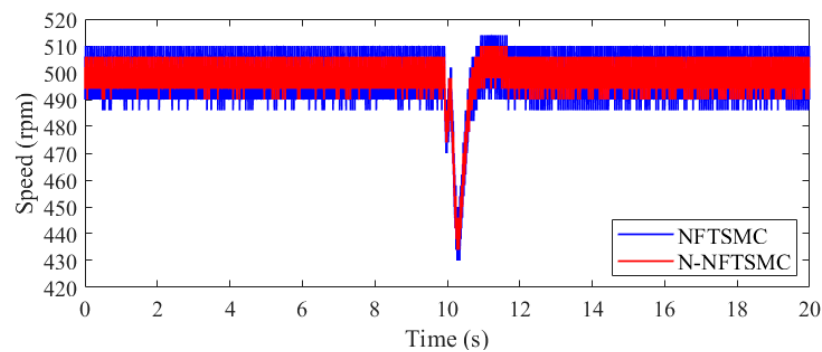


Figure 15. Experimental results at 500 rpm using a sudden load.

In order to further verify the robustness of N-NFTSMC, a loading experiment is conducted at a reference speed of 1000 rpm. From the experimental results in Figure 16, the chattering range of the N-NFTSMC is smaller than that of the NFTSMC. The N-NFTSMC also exhibits an anti-disturbance performance that is stronger than that of the NFTSMC.

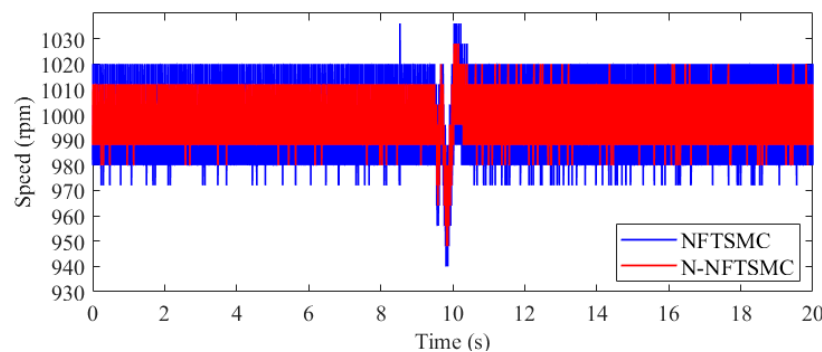


Figure 16. Experimental results at 1000 rpm using a sudden load.

The experimental results using different control methods are listed in Table 3 for clearer comparisons. We can see that, compared with the NFTSMC, the proposed N-NFTSMC shows smaller speed chattering. Moreover, when a sudden load is applied, the N-NFTSMC has a smaller maximum chattering value. These results verify that the N-NFTSMC can effectively suppress chattering and improve the anti-disturbance of the PMSM motion control system.

Table 3. Comparisons of experimental results using different control methods.

Control Method	Maximum Chattering Value (without Sudden Load)	Maximum Chattering Value (with Sudden Load)
NFTSMC (500 rpm)	10 rpm	70 rpm
N-NFTSMC (500 rpm)	6 rpm	66 rpm
NFTSMC (1000 rpm)	20 rpm	60 rpm
N-NFTSMC (1000 rpm)	12 rpm	52 rpm

7. Conclusions

In this work, we combined the nonsingular fast terminal sliding mode control with a sliding mode disturbance observer to enhance the chattering suppression and anti-disturbance capacity of the PMSM servo system. We conducted step response simulations that showed that our approach improves the response time and adequately suppresses chattering compared with the traditional FTSMC. Furthermore, we conducted sine tracking

simulations that showed that our approach has a lower maximum error compared with the traditional FTSMC.

In addition, we conducted the experiments using different speeds. The experimental results show that the N-NFTSMC improves the anti-disturbance performance and alleviates chattering. The results from the simulations and experiments indicate that the N-NFTSMC strengthens the servo system's dynamic performance and its disturbance mitigation. In the future, second-order sliding-mode algorithms will be considered for high-performance PMSM servo control.

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Nomenclature

Abbreviation	Full Name
PMSM	Permanent magnet synchronous motor
NFTSMC	Nonsingular fast terminal sliding mode control
N-NFTSMC	Novel nonsingular fast terminal sliding mode control
NSRL	New sliding mode reaching law
SDOB	Sliding mode disturbance observer
PID	Proportional–integral–derivative
ADRC	Active disturbance control
TVNDO	Time-varying nonlinear disturbance observer
ANNFTSMC	Adaptive neural network nonsingular fast terminal sliding mode control
IPMSM	Interior permanent magnet synchronous motor
SMC	Sliding mode control
NTSMC	Nonsingular terminal sliding mode control

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